

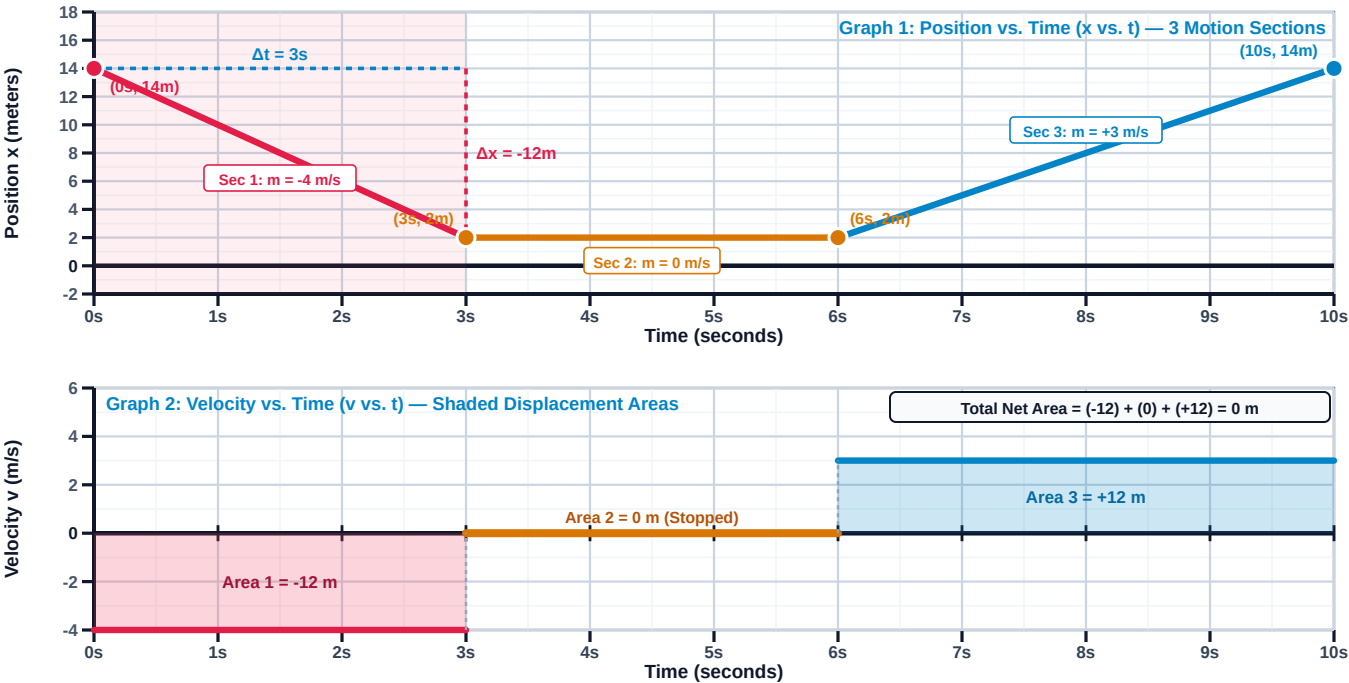
TEACHER MASTER KEY & EXEMPLAR • 3-STAGE DUAL-GRAPH STORY

TEACHER GRADING GUIDE • 10 POINTS

UNIT 2: KINEMATICS • DAY 15 PERFORMANCE TASK • DUAL-GRAPH STUDIO SCENARIO 1 EXEMPLAR

Criterion	Points	Mastery Indicators (Evidence of Full Credit)
1. Scenario & Narrative	2 pts	Original 3-stage story across 10.0s with clear object, reference origin (x=0m), realistic positions, and distinct stages (forward, stopped, or reverse).
2. 3-Stage Motion Table	3 pts	All 3 time intervals sum to 10.0s; correct signs on displacements (Δx); step-by-step GUESS velocity formulas ($v = \Delta x / \Delta t$) with correct units (m/s).
3. Graph 1 (x-t) & Slopes	2 pts	3 continuous connected line segments plotted accurately; slope calculations (Rise/Run) explicitly shown below graph matching table velocities.
4. Graph 2 (v-t) & Areas	2 pts	3 horizontal velocity bars drawn at correct heights; shaded rectangular regions between bars and v=0 axis; Area = $v \times \Delta t$ proves displacement Δx .
5. Dual-Graph Synthesis	1 pt	Sum of shaded areas matches net displacement ($x_{\text{final}} - x_{\text{initial}}$); correct physical meaning of negative slope/velocity; unit cancellation shown.

Worked Exemplar (Mars Rover Telemetry): Rover starts at $x = 14\text{m}$. Section 1 (0-3s): Reverses at -4 m/s to $x = 2\text{m}$ ($\Delta x = -12\text{m}$). Section 2 (3-6s): Stops for 3s to analyze rock sample ($\Delta x = 0\text{m}$). Section 3 (6-10s): Drives forward for 4s at $+3\text{ m/s}$ back to $x = 14\text{m}$ ($\Delta x = +12\text{m}$). Net $\Delta x = 0\text{m}$, Total Distance = 24m .



MATHEMATICAL VERIFICATION • SLOPE = VELOCITY & AREA = DISPLACEMENT

Section 1 (0s to 3s):

- x-t Slope: $(-12\text{m} - 0\text{m}) / (3\text{s}) = -4.0\text{ m/s}$
- v-t Area: $(-4\text{ m/s}) \times (3\text{ s}) = -12.0\text{ m}$
- Negative slope $\rightarrow$ negative velocity bar below axis!

Section 2 (3s to 6s):

- x-t Slope: $(0\text{m}) / (3\text{s}) = 0.0\text{ m/s}$
- v-t Area: $(0\text{ m/s}) \times (3\text{ s}) = 0.0\text{ m}$
- Flat horizontal position line $\rightarrow$ line rests on $v=0$ axis!

Section 3 (6s to 10s) & Total:

- x-t Slope: $(+12\text{m}) / (4\text{s}) = +3.0\text{ m/s}$
- v-t Area: $(+3\text{ m/s}) \times (4\text{ s}) = +12.0\text{ m}$
- Net Displacement: $\sum \text{Area} = -12\text{m} + 0\text{m} + 12\text{m} = 0.0\text{ m}$
- Matches Graph 1: $x_{\text{end}} (14\text{m}) - x_{\text{start}} (14\text{m}) = 0\text{ m}!$

DUAL-GRAPH ALIGNMENT PROOF